A Multi-Factor Memory-Based Affine Stochastic Volatility Model for Option Pricing and Mispricing Detection

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Abstract

We introduce a multi-factor memory-based stochastic volatility framework for option pricing. The model extends classical affine volatility models by incorporating multiple interacting memory factors that represent volatility dynamics across different time horizons. The resulting system preserves an affine structure, admits a tractable characteristic function representation, and allows efficient Fourier-based option pricing. The framework provides a mechanism for converting volatility forecasts into fair option values and ultimately into measurable option mispricing signals.

1 Introduction

Volatility exhibits several empirical properties that remain challenging for traditional option pricing models. These include volatility clustering, persistence across multiple time scales, implied volatility smiles, skew structures, and long-memory behavior.

Classical stochastic volatility models typically rely on a single latent variance factor. While such models often provide tractability, they may fail to capture the rich temporal structure observed in financial markets.

The objective of this framework is to construct a volatility process driven by multiple interacting memory components. Each component represents a different volatility horizon and contributes to the aggregate variance process. The resulting dynamics preserve affine tractability while introducing a flexible mechanism for modeling volatility persistence.

The model is designed to satisfy the following objectives:

(i) capture volatility clustering,

(ii) incorporate multiple memory scales,

(iii) preserve affine structure,

(iv) allow characteristic-function pricing,

(v) support FFT-based valuation,

(vi) generate fair-value estimates for option markets.

2 Model Construction

Let

$$S_t$$

denote the underlying asset price process.

Define the logarithmic price process

$$X_t = \log(S_t).$$

To model volatility persistence across different horizons, introduce a collection of memory factors

$$U_t^{(1)}, U_t^{(2)}, \dots, U_t^{(m)}.$$

Each factor represents a distinct component of the volatility environment.

3 Risk-Neutral Dynamics

Under the risk-neutral measure Q, the asset price evolves according to

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t,$$

where

- r denotes the risk-free rate,
- v_t denotes instantaneous variance,
- W_t is a Brownian motion under Q.

Applying Itô's formula yields the log-price dynamics

$$dX_t = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t}\,dW_t.$$

4 Variance Specification

The instantaneous variance is generated by the collection of memory factors.
Specifically,

$$v_t = \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)},$$

where

$$\omega > 0$$

represents the baseline variance level.
The coefficients

$$\alpha_i$$

determine the contribution of each memory factor to total variance.
Consequently, volatility is represented as a superposition of multiple memory horizons rather than a single latent factor.

5 Memory Dynamics

Each memory component evolves according to

$$dU_t^{(i)} = \left(-\lambda_i U_t^{(i)} + v_t\right) dt + \epsilon_i \sqrt{v_t}\, dZ_t^{(i)}.$$

The parameter

$$\lambda_i > 0$$

controls the rate at which memory decays.

Large values of λ_i correspond to short-term memory effects, whereas smaller values correspond to long-term persistence.

The coefficient

$$\epsilon_i$$

controls the volatility-of-volatility contribution associated with factor i.

6 Correlation Structure

For each factor, the driving Brownian motions satisfy

$$dW_t\, dZ_t^{(i)} = \rho_i\, dt,$$

with

$$-1 < \rho_i < 1.$$

Negative values of

$$\rho_i$$

generate asymmetric return-volatility interactions and produce implied-volatility skew structures commonly observed in equity markets.

7 Complete System

Combining the previous components yields the full model:

$$\boxed{\begin{aligned} dX_t &= \left(r - \frac{1}{2}v_t\right) dt + \sqrt{v_t}\, dW_t, \\ v_t &= \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)}, \\ dU_t^{(i)} &= \left(-\lambda_i U_t^{(i)} + v_t\right) dt + \epsilon_i \sqrt{v_t}\, dZ_t^{(i)}. \end{aligned}}$$

together with

$$dW_t\, dZ_t^{(i)} = \rho_i\, dt.$$

The above system forms the foundational structure of the proposed memory-based stochastic volatility framework.

8 Affine Structure

Define the state vector

$$Y_t = \left(X_t, U_t^{(1)}, U_t^{(2)}, \dots, U_t^{(m)}\right).$$

The proposed framework belongs to the class of affine stochastic processes.
To see this, observe that the variance process can be written as

$$v_t = \omega + \boldsymbol{\alpha}^\top \boldsymbol{U}_t,$$

where

$$\boldsymbol{U}_t = \begin{pmatrix} U_t^{(1)} \\ U_t^{(2)} \\ \vdots \\ U_t^{(m)} \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix}.$$

The drift of the memory vector becomes

$$d\boldsymbol{U}_t = (-\Lambda \boldsymbol{U}_t + v_t \mathbf{1})\, dt + \sqrt{v_t}\Sigma\, d\boldsymbol{Z}_t,$$

where

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_m \end{pmatrix},$$

and

$$\Sigma = \begin{pmatrix} \epsilon_1 & 0 & \cdots & 0 \\ 0 & \epsilon_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \epsilon_m \end{pmatrix}.$$

The drift is affine in the state variables.
Furthermore, the covariance matrix satisfies

$$d\langle Y \rangle_t = v_t A\, dt,$$

for an appropriate matrix A.

Since both drift and diffusion coefficients depend linearly upon the state variables, the process belongs to the affine class.

This property is critical because affine processes admit exponentially affine characteristic functions, allowing tractable option valuation.

9 Characteristic Function Representation

Define the conditional characteristic function

$$\phi(u, T) = \mathbb{E}^Q \left[e^{iuX_T} \mid \mathcal{F}_0 \right].$$

The affine property implies an exponential-affine representation of the form

$$\phi(u,T) = \exp\left(A(T,u) + \sum_{i=1}^{m} B_i(T,u)U_0^{(i)} + iuX_0\right).$$

Equivalently,

$$\phi(u,T) = \exp\left(A(T,u) + \mathbf{B}(T,u)^\top \mathbf{U}_0 + iuX_0\right).$$

The functions

$$A(T,u)$$

and

$$B_i(T,u)$$

fully characterize the distribution of future log-prices.

Consequently, option valuation reduces to solving a finite-dimensional system of ordinary differential equations.

10 Derivation of the Riccati System

Introduce

$$\kappa(u) = \frac{1}{2}\left(u^2 + iu\right).$$

Define

$$G(T,u) = -\kappa(u) + \sum_{j=1}^{m} B_j(T,u).$$

Substituting the exponential-affine ansatz into the Kolmogorov backward equation yields a coupled Riccati system.

For each memory factor,

$$\frac{dB_i}{dT} = -\lambda_i B_i + \alpha_i G + \frac{1}{2}\epsilon_i^2 B_i^2 + \rho_i \epsilon_i u B_i.$$

The first term

$$-\lambda_i B_i$$

represents mean reversion.

The second term

$$\alpha_i G$$

captures the contribution of factor i to total variance.

The nonlinear term

$$\frac{1}{2}\epsilon_i^2 B_i^2$$

is responsible for volatility-of-volatility effects and moment explosions.

The correlation contribution

$$\rho_i \epsilon_i u B_i$$

generates skew dynamics.

The equation for the scalar coefficient $A(T,u)$ becomes

$$\frac{dA}{dT} = iur + \omega G.$$

Initial conditions are

$$A(0,u) = 0,$$

and

$$B_i(0,u) = 0, \qquad i = 1, \ldots, m.$$

The solution of this Riccati system completely determines the characteristic function.

11 European Option Valuation

Consider a European call option with strike K and maturity T.

Its risk-neutral value is

$$C(K,T) = e^{-rT}\mathbb{E}^Q\left[(S_T - K)^+\right].$$

Using Fourier-transform techniques, the payoff can be represented in frequency space.

Substituting the characteristic function yields

$$C(K,T) = e^{-rT}\frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\phi(u-i,T)}{iu\,\phi(-i,T)}\right] du.$$

All information regarding volatility, skew, memory effects, and higher moments enters exclusively through

$$\phi(u,T).$$

Thus, once the Riccati system has been solved, option prices can be obtained through numerical integration.

12 Fast Fourier Transform Pricing

Direct numerical integration for every strike is computationally expensive.

Suppose a volatility surface contains

$$N$$

strikes.

A naïve implementation requires

$$O(N^2)$$

operations.

To improve efficiency, evaluate

$$\phi(u,T)$$

on a regularly spaced frequency grid

$$u_1, u_2, \ldots, u_N.$$

The Carr–Madan Fourier formulation permits simultaneous computation of all strikes through the Fast Fourier Transform.

The computational complexity becomes

$$O(N \log N).$$

This reduction is crucial during calibration because thousands of option prices must be evaluated repeatedly.

Consequently, the affine-memory framework remains computationally feasible even when multiple volatility factors are included.

13 Calibration Framework

The practical usefulness of the model depends upon the ability to infer parameter values from observed option prices.

Define the parameter vector

$$\theta = (\omega, \alpha_1, \ldots, \alpha_m, \lambda_1, \ldots, \lambda_m, \epsilon_1, \ldots, \epsilon_m, \rho_1, \ldots, \rho_m).$$

Suppose market observations consist of implied volatilities

$$\sigma_1^{\text{market}}, \sigma_2^{\text{market}}, \ldots, \sigma_N^{\text{market}}.$$

For a given parameter vector θ, the model generates

$$\sigma_1^{\text{model}}(\theta), \sigma_2^{\text{model}}(\theta), \ldots, \sigma_N^{\text{model}}(\theta).$$

Calibration is formulated as a nonlinear least-squares optimization problem.

Define the loss function

$$L(\theta) = \sum_{i=1}^{N} w_i \left(\sigma_i^{\text{model}} - \sigma_i^{\text{market}}\right)^2,$$

where w_i denotes a weighting coefficient.

Typical choices include:

$$w_i = 1,$$

uniform weighting,

$$w_i = \frac{1}{\text{BidAskSpread}_i},$$

liquidity weighting,

or

$$w_i = \frac{1}{\text{Vega}_i^2},$$

which prevents highly sensitive options from dominating the optimization.

The calibrated parameter vector is

$$\theta^* = \operatorname*{argmin}_{\theta} L(\theta).$$

The resulting parameter set generates the volatility surface that most closely matches observed market prices.

14 Sensitivity Analysis and Jacobian Structure

Define the residual vector

$$r_i = \sigma_i^{\text{model}} - \sigma_i^{\text{market}}.$$

Collecting all residuals yields

$$r(\theta) = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{pmatrix}.$$

The Jacobian matrix is defined by

$$J_{ij} = \frac{\partial r_i}{\partial \theta_j}.$$

The Jacobian measures the sensitivity of each point on the volatility surface with respect to each model parameter.

Because the characteristic function admits the affine representation

$$\phi(u, T) = \exp\Big(A(T, u) + \mathbf{B}(T, u)^\top \mathbf{U}_0 + iuX_0\Big),$$

parameter derivatives satisfy

$$\frac{\partial \phi}{\partial \theta} = \phi \left(\frac{\partial A}{\partial \theta} + \mathbf{U}_0^\top \frac{\partial \mathbf{B}}{\partial \theta} \right).$$

Differentiating the Riccati system produces a secondary system of sensitivity equations.

Consequently, all Jacobian entries can be computed analytically while solving the Riccati equations.

This substantially accelerates calibration compared with finite-difference methods.

15 Gauss–Newton Approximation

The loss function can be written as

$$L(\theta) = r(\theta)^\top r(\theta).$$

Its gradient is

$$\nabla L(\theta) = 2J^\top r.$$

The exact Hessian is

$$H = 2J^\top J + 2\sum_{i=1}^{N} r_i \nabla^2 r_i.$$

Computing the second term is expensive.

Near the optimum the residuals are small, yielding the approximation

$$H \approx 2J^\top J.$$

This is the Gauss–Newton approximation.

The resulting parameter update becomes

$$\theta_{n+1} = \theta_n - \left(J^\top J\right)^{-1} J^\top r.$$

When calibration is already close to the optimum, convergence is often nearly quadratic.

16 Levenberg–Marquardt Regularization

Far from the optimum the matrix

$$J^\top J$$

may become nearly singular.
To stabilize optimization one introduces a damping parameter

$$\eta > 0.$$

The modified Hessian approximation becomes

$$H_{\text{LM}} = J^\top J + \eta I.$$

The parameter update becomes

$$\theta_{n+1} = \theta_n - \left(J^\top J + \eta I\right)^{-1} J^\top r.$$

For large values of η,

$$H_{\text{LM}} \approx \eta I,$$

and the method behaves similarly to gradient descent.
For small values of η,

$$H_{\text{LM}} \approx J^\top J,$$

recovering Gauss–Newton.
The Levenberg–Marquardt method therefore combines stability with rapid convergence.

17 Short-Maturity Skew Asymptotics

Define log-moneyness

$$k = \log\left(\frac{K}{S_0}\right).$$

The implied-volatility skew is

$$\frac{\partial \sigma_{\text{imp}}}{\partial k}.$$

The at-the-money skew is

$$\left.\frac{\partial \sigma_{\text{imp}}}{\partial k}\right|_{k=0}.$$

A first-order asymptotic expansion yields

$$\left.\frac{\partial \sigma_{\text{imp}}}{\partial k}\right|_{k=0} \approx \frac{1}{2\sqrt{v_0}} \sum_{i=1}^{m} \alpha_i \epsilon_i \rho_i.$$

Several conclusions follow:

1. Larger volatility-of-volatility parameters increase skew magnitude.
2. Negative correlations generate negative equity skew.
3. Multiple memory factors contribute additively.
4. Long-memory factors influence skew persistence.

The skew structure is therefore jointly controlled by

$$\alpha_i, \qquad \epsilon_i, \qquad \rho_i.$$

18 Smile Curvature

The curvature of the implied-volatility smile is measured by

$$\left.\frac{\partial^2 \sigma_{\text{imp}}}{\partial k^2}\right|_{k=0}.$$

Within the affine-memory framework,

$$\left.\frac{\partial^2 \sigma_{\text{imp}}}{\partial k^2}\right|_{k=0} \propto \sum_{i=1}^{m} \alpha_i \epsilon_i^2.$$

This relationship highlights the role of volatility-of-volatility in generating smile curvature. Hence:

- larger ϵ_i creates steeper smiles,
- additional factors increase shape flexibility,
- curvature remains stable under affine dynamics.

The model can therefore simultaneously fit short-dated smiles and long-dated smirks.

19 Wing Asymptotics and Moment Explosions

Extreme-strike behavior is governed by the nonlinear Riccati term

$$\frac{1}{2}\epsilon_i^2 B_i^2.$$

This term controls the existence of higher moments.
Moment explosions occur whenever the Riccati solution becomes singular at finite maturity.
Define the largest finite moment

$$p^+(T).$$

Applying Lee's moment formula gives

$$\lim_{k\to\infty} \frac{w(k,T)}{k} = \frac{2}{p^+(T)},$$

where

$$w(k,T) = \sigma_{\text{imp}}^2(k,T)\,T$$

denotes total implied variance.

Similarly, the left wing is governed by

$$p^-(T).$$

Consequently:

- ϵ_i governs tail thickness,
- ρ_i influences asymmetry,
- memory factors alter long-horizon tail behavior.

These parameters therefore directly affect deep out-of-the-money option prices.

20 Economic Interpretation

The variance process admits the decomposition

$$v_t = \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)}.$$

The constant component

$$\omega$$

represents the long-run volatility floor.
The memory factors represent volatility operating across multiple horizons.
For example,

$$(\lambda_1, \lambda_2, \lambda_3) = (10, 2, 0.2)$$

may correspond to

$$\begin{aligned} \text{Factor 1}: &\quad \text{daily memory,} \\ \text{Factor 2}: &\quad \text{weekly memory,} \\ \text{Factor 3}: &\quad \text{monthly memory.} \end{aligned}$$

The variance process may therefore be viewed as

$$\text{Variance} = \text{Baseline} + \text{Short-Term Memory} + \text{Medium-Term Memory} + \text{Long-Term Memory}.$$

Unlike classical one-factor models, the framework explicitly separates volatility persistence across multiple time scales.

21 Mispricing Signal Framework

The characteristic function generates a complete option-pricing engine.
Given

$$\left(U_t^{(1)}, \ldots, U_t^{(m)}\right),$$

the model produces a theoretical value

$$C_{\text{model}}(K, T).$$

Observed market prices provide

$$C_{\text{market}}(K, T).$$

Define the pricing discrepancy

$$\Delta C = C_{\text{market}} - C_{\text{model}}.$$

Interpretation is immediate:

$$\Delta C > 0$$

implies the option is expensive relative to the model,
while

$$\Delta C < 0$$

implies the option is cheap relative to the model.
To normalize signals across strikes and maturities define

$$Z = \frac{C_{\text{market}} - C_{\text{model}}}{\sigma_\Delta}.$$

A practical trading rule may be

$$|Z| > 2$$

for signal generation,
and

$$|Z| > 3$$

for strong conviction opportunities.
The pricing model itself does not generate alpha.
Instead:

$$\text{Volatility Forecast} \rightarrow \text{Fair Option Value} \rightarrow \text{Mispricing Signal.}$$

The forecasting engine predicts future volatility states.
The affine-memory pricing model converts those forecasts into option values.
The discrepancy between theoretical value and market value creates the trading signal.

22 Indian Market Extension

The baseline affine-memory volatility framework is sufficiently general to model option markets in any jurisdiction.

However, index-option markets in India exhibit several distinctive features that are not fully captured by classical stochastic-volatility models.

These include:

1. Weekly expiry effects,
2. Open-interest concentration,
3. Put-call ratio imbalances,
4. Event-driven volatility shocks associated with RBI policy announcements, Union Budgets, and elections.

To incorporate these features, we extend the variance specification.

22.1 Risk-Neutral Dynamics with Dividend Yield

For Indian index options, the log-price process is written as

$$dX_t = \left(r_t - q_t - \frac{1}{2}v_t\right) dt + \sqrt{v_t}\, dW_t,$$

where

$$r_t$$

denotes the domestic risk-free rate term structure and

$$q_t$$

denotes the dividend yield associated with the index.

22.2 Expiry Effect

Indian index-option markets exhibit pronounced volatility distortions during weekly expiry periods.

To capture this phenomenon, define an expiry-state variable

$$E_t = \exp(-\gamma\tau_t),$$

where

$$\tau_t$$

denotes time remaining until expiry.

The variance process becomes

$$v_t = \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)} + \eta E_t.$$

The coefficient

$$\eta$$

measures the strength of expiry-related volatility pressure.

As expiry approaches,

$$\tau_t \to 0,$$

implying

$$E_t \to 1.$$

Consequently, the model generates elevated short-term volatility consistent with observed market behavior.

22.3 Open Interest and Put–Call Ratio Effects

Option-market positioning contains information regarding market expectations and dealer inventory pressures.

Define

$$OI_t$$

as a normalized open-interest factor and

$$PCR_t$$

as the put-call ratio factor.

The variance specification becomes

$$v_t = \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)} + \beta OI_t + \gamma PCR_t + \eta E_t.$$

The parameter

$$\beta$$

measures the impact of option positioning.

Similarly,

$$\gamma$$

captures directional pressure reflected in the put-call ratio.

These variables introduce market microstructure information directly into the volatility process.

22.4 Event-Driven Jump Component

Indian financial markets frequently experience abrupt volatility shocks associated with macroeconomic and political events.

Examples include:

- RBI policy announcements,
- Union Budget releases,
- National elections,
- Major geopolitical developments.

To incorporate these effects, augment the log-price dynamics with a jump process

$$dX_t = \left(r_t - q_t - \frac{1}{2} v_t \right) dt + \sqrt{v_t}\, dW_t + J_t\, dN_t,$$

where

$$N_t$$

is a Poisson process and

$$J_t$$

denotes the jump magnitude.

The resulting system combines memory-driven volatility with event-driven discontinuities.

22.5 Extended Indian Affine-Memory Model

Combining the previous components yields

$$\boxed{\begin{aligned} dX_t &= \left(r_t - q_t - \frac{1}{2}v_t\right) dt + \sqrt{v_t}\, dW_t + J_t\, dN_t, \\ v_t &= \omega + \sum_{i=1}^{m} \alpha_i U_t^{(i)} + \beta OI_t + \gamma PCR_t + \eta E_t, \\ dU_t^{(i)} &= \left(-\lambda_i U_t^{(i)} + v_t\right) dt + \epsilon_i \sqrt{v_t}\, dZ_t^{(i)}. \end{aligned}}$$

22.6 Extended Characteristic Function

Under the affine specification, the characteristic function retains the exponential-affine representation

$$\phi(u,T) = \exp\left(A(T,u) + \sum_{i=1}^{m} B_i(T,u)U_0^{(i)} + C(T,u)OI_0 + D(T,u)PCR_0 + E(T,u)E_0 + iuX_0\right).$$

Consequently, the pricing problem remains tractable and reduces to solving an enlarged system of Riccati equations.

22.7 Economic Interpretation

The extended framework combines four distinct sources of information:

$$\text{Memory Dynamics} + \text{Market Positioning} + \text{Expiry Effects} + \text{Event Risk}.$$

Unlike classical Heston-type models, the framework incorporates observable option-market structure directly into the variance process.

As a result, the model is capable of representing volatility patterns commonly observed in Indian index-option markets while preserving the computational advantages of affine stochastic-volatility models.

23 Conclusion

A multi-factor memory-based affine stochastic volatility framework has been developed for option pricing and volatility-surface modeling.

The model combines multiple memory horizons with affine tractability and produces a characteristic-function representation governed by a finite-dimensional Riccati system.

The framework supports:

- Fourier pricing,
- FFT acceleration,
- efficient calibration,
- skew generation,
- smile curvature control,
- realistic tail behavior,

- multi-horizon volatility persistence.

The resulting structure establishes a direct mathematical bridge between volatility forecasting, option valuation, and systematic mispricing detection.